Tamminana Amrutha

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.69
NARAYANA HIGH SCHOOL CHINAGADILI VIZAG AP	Class XII (CENTRAL BOARD OF SECONDARY EDUCATION)	2024	95.60%
NARAYANA ENG MEDIUM SCH PARADESIPALEM VIZAG AP	Class X (CENTRAL BOARD OF SECONDARY EDUCATION)	2022	97.00%

SKILLS

- **Programming Languages:** C++, C, SQL, RISC-V Assembly, Java*, Python*
- **Technical Tools:** Solid Edge, Linux, LaTeX*, Shell Scripting*, PostgreSQL*
- **Core Computer Science:** Data Structures, Algorithms, Operating Systems, Database Management Systems, Computer Architecture
- **Soft skills:** Team management, Adaptability, Communication, Time management

PROJECTS

Kernel Extensions for xv6

- Extended the xv6 operating system by implementing kernel-level features across process management, scheduling, virtual memory, and storage subsystems.
- Designed and implemented a **System-Call-Aware Multi-Level Feedback Queue (SC-MLFQ)** scheduler with four priority queues, queue-specific time quanta, dynamic priority management, and per-process scheduling statistics.
- Enhanced the virtual memory subsystem by implementing a global frame table, a **Clock page-replacement algorithm**, and a disk-backed swap framework integrated with scheduler priorities.
- Implemented a software RAID storage layer supporting **RAID 0, RAID 1**, and **RAID 5**, along with FCFS and SSTF disk scheduling policies and kernel-level I/O statistics.

Performance Analysis of ALU-FPU Parallelism on ARM Cortex-A76

- Investigated the effectiveness of **ALU-FPU parallel execution** by developing and experimentally evaluating alternative implementations across **seven compute-intensive** workloads on ARM Cortex-A76.
- Evaluated multiple optimization techniques like loop transformations, accumulator-based restructuring, and dependency breaking to improve execution time across diverse workloads.
- Identified key performance factors such as instruction dependencies, memory locality, compiler auto-vectorization, and integer-floating-point conversion overhead, achieving around **50-60%** execution-time improvement across several workloads.

PostgreSQL Indexing & Query Performance Analysis

- Analyzed the impact of **indexing and data loading order** on query performance, reducing overall query latency by ~23% by building indexes after bulk data insertion.
- Developed a method to estimate threshold values for target **selectivity** and validated the results through experiments.
- Benchmarked range queries across **51 selectivity levels** to compare the performance of indexes, sequential scans, and clustered data, and analyzed how different indexing and data access methods affect database performance.

Arbitrary Precision Arithmetic Library

- Engineered custom **AInteger** and **AFloat** classes from scratch to support arbitrary-precision arithmetic using

string-based representations instead of native numeric data types.
- Designed and implemented manual digit-wise algorithms for addition, subtraction, multiplication, and division with carry propagation, borrow handling, normalization, and decimal-point management.
- Implemented floating-point arithmetic with precision up to 30 decimal places by leveraging underlying integer arithmetic and custom decimal-point management.

RELEVANT COURSES

CSE core: Data Structures, Algorithms, Operating Systems, Database Management Systems, Computer Architecture, Software Development Fundamentals, Theory of Computation, Artificial intelligence, Discrete Mathematics, Compilers-1, Introduction to Programming.

Mathematics and EE Core: Probability & Random Variables, Linear Algebra, Digital Circuits, Statistics, Calculus, Complex Variables, Transform Techniques

POSITIONS OF RESPONSIBILITY

Event Coordinator, Indian Knowledge System Student Chapter (IKS-SC):

- Coordinated planning and execution activities for events with 200+ participants, contributing to logistics, scheduling, stakeholder coordination, and on-site event operations.
- Coordinated logistics and execution of educational outreach programs, including transportation coordination and structured participant engagement sessions.

EXTRACURRICULAR

- Trained in classical vocal music for 5+ years, performing at temple gatherings and cultural events, strengthening public speaking confidence and stage presence.
- Vocalist, Vibes Club, IIT Hyderabad (Aug 2024 - May 2025).
- Actively engaged in competitive programming through regular participation on LeetCode and Codeforces.